

StatFlow AI

Automated Statistical & Machine Learning Analysis Report

Statistical Analysis • Machine Learning • AI Business Insights

Generated automatically from the uploaded dataset

Executive Summary

CLEAN OBSERVATIONS 238	ML MODEL Multiple Linear Regression	R² 0.1986
MAE 999.97	RMSE 1308.62	AVERAGE PROFIT -401.50

AI-Generated Analysis

Key Findings

Metric	Value
Total observations	238
Training set	190
Testing set	48
Average profit	-\$401.5 (median -\$350.55)
Profit volatility	$\sigma = \$1,470.35$ (min -\$5,187.47, max \$4,167.08)
Revenue-Profit correlation	0.438 (moderate positive)
Advertising-Profit correlation	-0.37 (moderate negative)
Discount-Profit correlation	-0.108 (weak negative)
Unit-Price-Profit correlation	0.329 (weak positive)
Customer-Satisfaction-Profit correlation	0.329 (same magnitude as Unit-Price)
Model R-squared	0.199 (explains ~20 % of profit variance)
MAE / RMSE	\$999.97 / \$1,308.62 (large prediction errors)

Strongest Correlations

Pair	Correlation	Interpretation
Unit_Price vs Revenue	0.69	Higher price tends to generate higher revenue (strongest linear link in the data).
Units_Sold vs Revenue	0.504	More units sold also lifts revenue, but less strongly than price.
Revenue vs Profit	0.438	Revenue growth modestly improves profit, but profit is still often negative.
Advertising_Spend vs Profit	-0.37	More ad spend is associated with lower profit—potentially inefficient spend.
Unit_Price vs Profit	0.329	Higher price modestly helps profit.
Customer_Satisfaction vs Profit	0.329	Satisfaction also shows a modest positive link to profit.

Business Insights

1. Revenue is driven primarily by price and volume.

- A 0.69 correlation between Unit_Price and Revenue indicates that price adjustments have a pronounced effect on top-line sales.
- Units_Sold also contributes positively (0.504), suggesting that volume and price work together to lift revenue.

1. Profitability is fragile.

- The average profit is slightly negative (-\$401.5) with a high standard deviation, meaning many transactions lose money while a few generate large gains.
- Only ~20 % of profit variation is captured by the regression model, indicating other unobserved factors (cost structure, overhead, product mix) dominate.

1. Discounting erodes profit.

- Discount shows a weak negative correlation with Revenue (-0.108) and no direct profit coefficient is provided, but the model's **Discount coefficient = -1,650.71** signals that each 0.01 increase in discount (i.e., 1 % discount) cuts predicted profit by roughly \$1,650, all else equal.
- Reducing discount depth could improve margins.

1. Advertising spend appears counter-productive.

- The negative correlation (-0.37) and a coefficient of **-0.04276** suggest that each additional dollar of ad spend reduces predicted profit by about 4 cents, implying diminishing returns or mis-targeted campaigns.

1. Customer satisfaction has a modest upside.

- A coefficient of **+499.84** means that a one-point rise in satisfaction (on the scale used) adds roughly \$500 to predicted profit, holding other variables constant.
- Investing in experience improvements could yield measurable profit gains.

1. Model limitations.

- MAE ≈ \$1,000 and RMSE ≈ \$1,309 are large relative to the profit range, and the sample predictions frequently miss the actual sign (e.g., predicted profit often negative while actual profit is positive).
- R² of 0.199 confirms the model is a **poor predictor** of profit; it should be used only for directional insight, not precise forecasting.

Risks

Risk	Why it matters
Persistent negative average profit	Ongoing losses can erode cash reserves and limit growth.
High profit volatility	Large swings make budgeting and cash-flow planning unreliable.
Inefficient advertising spend	Money spent on ads may not translate into profit, inflating cost base.
Over-reliance on discounting	Deep discounts can cannibalize margins without delivering sufficient volume lift.
Limited explanatory power of the predictive model	Decisions based on the current model may be mis-guided because it captures only a small fraction of profit drivers.
Potential data quality issues	The wide profit range (-\$5,187 to \$4,167) and high standard deviation suggest outliers or heterogeneous product mixes that could distort insights.

Recommendations

1. Re-evaluate Pricing Strategy

- Conduct price elasticity testing to pinpoint the optimal price point that maximizes revenue *and* profit.
- Avoid blanket price cuts; instead, target price adjustments on high-margin SKUs.

1. Trim or Restructure Discount Policy

- Quantify the profit impact of each discount tier; consider moving from linear discounts to promotional bundles that preserve margin.
- Set a maximum discount ceiling (e.g., 10 % of list price) to protect profitability.

1. Optimize Advertising Spend

- Perform a ROI analysis on each advertising channel; shift budget toward channels with positive profit correlation.
- Adopt performance-based models (e.g., pay-per-click with conversion tracking) to ensure spend directly ties to revenue generation.

1. Invest in Customer Satisfaction Initiatives

- Prioritize improvements that have the highest impact on satisfaction scores (e.g., faster fulfillment, responsive support).
- Use A/B tests to measure the profit lift from satisfaction-driven changes.

1. Refine the Profit-Prediction Model

- Augment the feature set with cost-of-goods-sold, fixed overhead, and product-mix variables to capture more profit drivers.
- Collect additional data (e.g., repeat purchase rate, churn) to improve model R^2 .
- Validate predictions on a larger hold-out set before using them for strategic decisions.

1. Monitor Profitability Metrics in Real Time

- Implement dashboards that track profit per transaction, margin by product line, and advertising ROI.
- Set alerts when profit falls below a pre-defined threshold, enabling rapid corrective action.

1. Conduct Periodic Profitability Audits

- Review high-profit outliers to understand best-practice drivers (e.g., product mix, low-cost fulfillment).
- Replicate successful patterns across the portfolio while mitigating the impact of low-margin transactions.

1. Consider Segmentation

- Segment customers or products by profitability rather than aggregating all transactions.
 - Tailor pricing, discount, and marketing tactics to each segment to maximize overall profit.
-

Bottom Line: The data shows that revenue can be boosted by price and volume, but profit is currently marginal and volatile. Discounting and advertising appear to be the biggest profit drains, while customer satisfaction offers a modest upside. Because the predictive model explains only ~20 % of profit variation, any strategic move should be backed by targeted experiments and continuous profitability monitoring rather than relying solely on the current regression output. By tightening discount policies, reallocating ad spend, and focusing on high-satisfaction customer segments, the business can move toward a more sustainable profit trajectory.
